

ML Graded Quiz M2

Question 1

Read the swap rate data below:

```
! pip install pandas-datareader

import pandas_datareader.data as web

from datetime import datetime

start = datetime(2002, 6, 30)

end = datetime(2022, 6, 30)

swap_data = ['DSWP1', 'DSWP2', 'DSWP3', 'DSWP4', 'DSWP5']

swap_df = web.DataReader(swap_data, 'fred', start, end)
```

After doing the initial steps of detrending the data and computing the covariance matrix, what is the total variance of the covariance matrix?

- ☐ 8.0
- ☐ 7.6
- ☐ 7.9
- ☒ 7.5

Question 2

Which of the following is **false** about agglomerative hierarchical clustering?

- ☐ We must retain clusters formed in the previous iteration.
- ☐ Nearly all the time we will be able to identify the number of clusters by studying the dendrogram.
- ☐ It is computationally heavy and would require a lot of memory for larger datasets.
- ☒ It might be difficult to identify the number of clusters in a dataset and we might need assistance from a domain expert.

Question 3

Why would we use PCA for clustering?

- ☐ Finding data that is already transformed after centering and scaling
- ☐ Finding the best explained variance
- ☐ Including additional features to our dataset
- ☒ Improving the cluster performance by finding new features

Question 4

What is the difference between supervised learning and unsupervised learning?

- ☐ Unsupervised learning adds patterns to the data while supervised learning adds patterns to an existing structure.
- ☐ Unsupervised learning is unguided by a prediction task while supervised learning is guided by a prediction task.
- ☐ Unsupervised learning cannot be generalized on unseen data while supervised learning is easily extended to test data.
- ☒ Unsupervised learning is used in classification and dimensionality reduction problems while supervised learning is used in regression problems.



QUESTION 1



Question 5

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swap_data = ['DSWP1', 'DSWP2', 'DSWP3', 'DSWP4', 'DSWP5',
             'DSWP6', 'DSWP7', 'DSWP8', 'DSWP9', 'DSWP10',
             'DSWP11', 'DSWP12', 'DSWP13', 'DSWP14', 'DSWP15',
             'DSWP16', 'DSWP17', 'DSWP18', 'DSWP19', 'DSWP20',
             'DSWP21', 'DSWP22', 'DSWP23', 'DSWP24', 'DSWP25',
             'DSWP26', 'DSWP27', 'DSWP28', 'DSWP29', 'DSWP30',
             'DSWP31', 'DSWP32', 'DSWP33', 'DSWP34', 'DSWP35',
             'DSWP36', 'DSWP37', 'DSWP38', 'DSWP39', 'DSWP40',
             'DSWP41', 'DSWP42', 'DSWP43', 'DSWP44', 'DSWP45',
             'DSWP46', 'DSWP47', 'DSWP48', 'DSWP49', 'DSWP50',
             'DSWP51', 'DSWP52', 'DSWP53', 'DSWP54', 'DSWP55',
             'DSWP56', 'DSWP57', 'DSWP58', 'DSWP59', 'DSWP60',
             'DSWP61', 'DSWP62', 'DSWP63', 'DSWP64', 'DSWP65',
             'DSWP66', 'DSWP67', 'DSWP68', 'DSWP69', 'DSWP70',
             'DSWP71', 'DSWP72', 'DSWP73', 'DSWP74', 'DSWP75',
             'DSWP76', 'DSWP77', 'DSWP78', 'DSWP79', 'DSWP80',
             'DSWP81', 'DSWP82', 'DSWP83', 'DSWP84', 'DSWP85',
             'DSWP86', 'DSWP87', 'DSWP88', 'DSWP89', 'DSWP90',
             'DSWP91', 'DSWP92', 'DSWP93', 'DSWP94', 'DSWP95',
             'DSWP96', 'DSWP97', 'DSWP98', 'DSWP99', 'DSWP100']

swap_df = web.DataReader(swap_data, 'fred', start, end)
```

After doing the initial steps of detrending the data and computing the covariance matrix, calculate the exposure brought about by the first loading if we have a portfolio with the exposures to interest rates variation as shown in the table below.

2 years rate	3 years rate	5 years rate	7 years rate	10 years rate
10	5	-9	-8	2

(DWSXP represents X yearly rates)

- ☐ 0.26706
- ☐ 0.127044
- ☐ 0.27165
- ☒ -3.15854

Question 6

Find the Canberra distance between the points (1,3), (2,8).

- ☐ 5.100
- ☐ 0.788
- ☐ 0.682
- ☒ 5

Question 7

Which of the following is **not** true about PCAs?

- ☐ The number of Principal components can exceed the number of original features
- ☐ It is a linear transformation technique
- ☐ They consider direction of the data points
- ☒ They consider the direction of maximum variability

Question 8

How do we use the elbow method in k-means?

- ☐ Get the number of clusters.
- ☐ Minimize the cost function.
- ☐ Get the optimal points we can use in a cluster.
- ☒ Minimize randomization of the initial points.

Question 9

Why do we standardize the features on the same scale before applying principal component analysis?

- ☐ To distinguish principal components better
- ☐ To prevent features with large outliers from dominating the outcome
- ☐ To improve interpretability of the principal components
- ☒ To increase visibility of the correlated features

Question 10

Which of the following is used to store the distances between each point?

- ☐ Symmetric matrix
- ☐ Primary matrix
- ☐ Proximity matrix
- ☒ Euclidean matrix

Question 11

If we had 10 features in our dataset and we performed PCA on the dataset, how many principal components will we have?

- ☐ 10
- ☐ The first n that will give us a cumulative variance of at least 80%
- ☐ The first n that will give us a cumulative variance of at most 80%
- ☒ 3

Question 12

Which of the following statements about k-means is **true**?

- ☐ New points are assigned cluster membership based on existing clusters.
- ☐ The objective function of k-means is the mean of the distances of the cluster centers to all the points in the clusters.
- ☐ It finds a local minimum of maximizing squared distances.
- ☒ Centroids are fixed points chosen prior to the clustering process.

Question 13

Download the swap rate data below:

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```

After doing the initial steps of detrending the data and computing the covariance matrix, what is the covariance of DWSP3 and DWSP4?

- ☐ 0.9924
- ☐ 0.9953
- ☐ 0.9660
- ☒ 0.9896

Question 14

We can use hierarchical clustering for which of the following applications?

- ☐ Detecting Outliers
- ☐ Selecting features in a supervised learning problem
- ☐ Imputing missing values to their cluster mean
- ☒ Recognizing speech patterns

Question 15

What are the eigenvalues of the covariance matrix equivalent to?

- ☐ Variance
- ☐ Correlation
- ☐ Standard deviation
- ☒ Mean

Question 16

Which of the following is **not** an application of Gaussian mixture models?

- ☐ Clustering
- ☐ Creating non-parametric density models

- ☐ Feature extraction
- ☒ Testing how likely a new point would belong to a group

SUBMIT

